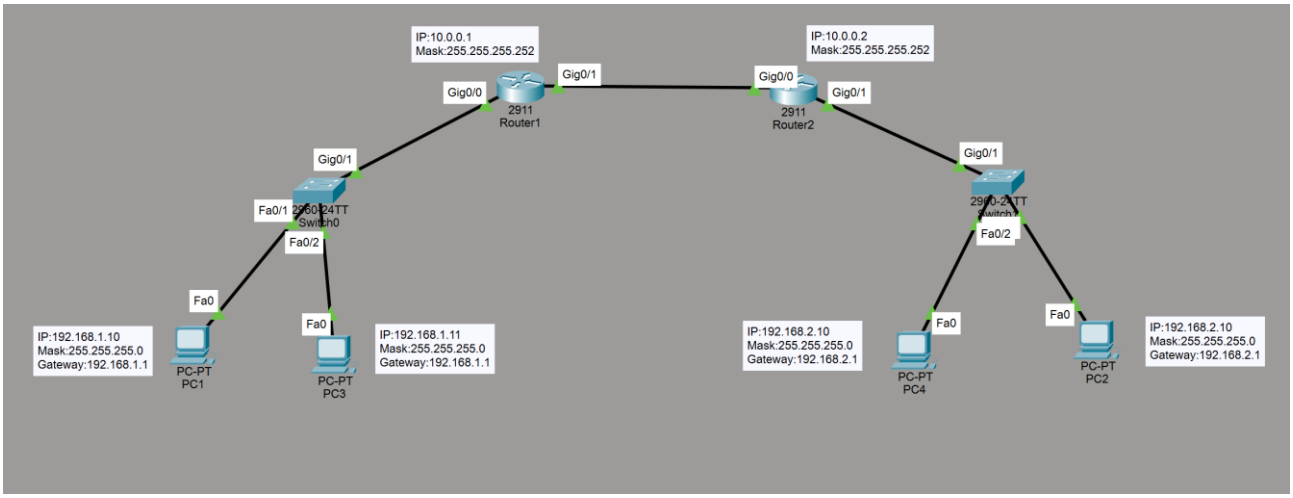
ACL Lab

Topology:



Addressing:

Device	IP Address	Subnet Mask	Default Gateway
PC1	192.168.1.10	255.255.255.0	192.168.1.1
PC3	192.168.1.11	255.255.255.0	192.168.1.1
Router1	192.168.1.1	255.255.255.0	
Router1 to Router2	10.0.0.1	255.255.255.252	
Router2 to Router1	10.0.0.2	255.255.255.252	
Router2	192.168.2.1	255.255.255.0	
PC2	192.168.2.10	255.255.255.0	192.168.2.1
PC4	192.168.2.11	255.255.255.0	192.168.2.1

Connectivity Verification:

Ping from PC 1 to PC2:

```
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Stage 1 — Basic Standard ACL

Objective:

Block traffic only from PC1.

R1:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#access-list 10 deny host 192.168.1.10
Router(config)#access-list 10 permit any
Router(config)#interface g0/1
Router(config-if)#ip access-group 10 out
Router(config-if)#
```

Result:

```
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Stage 2 — Subnet Filtering

Objective:

Block the entire 192.168.1.0/24 network.

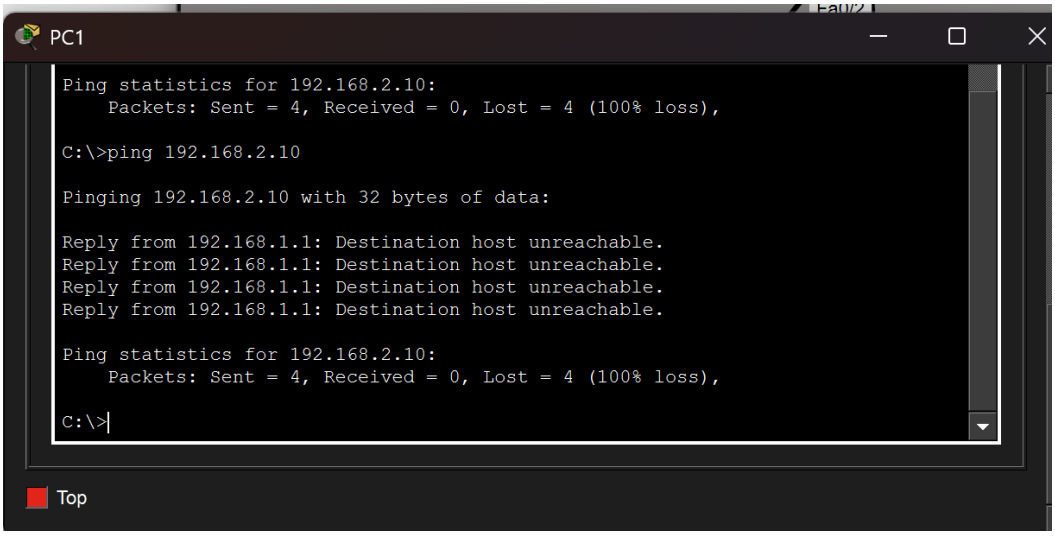
R1:

```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#no access-list 10
Router(config)#access-list 10 deny 192.168.1.0 0.0.0.255
Router(config)#acess-list 10 permit any
                ^
% Invalid input detected at '^' marker.

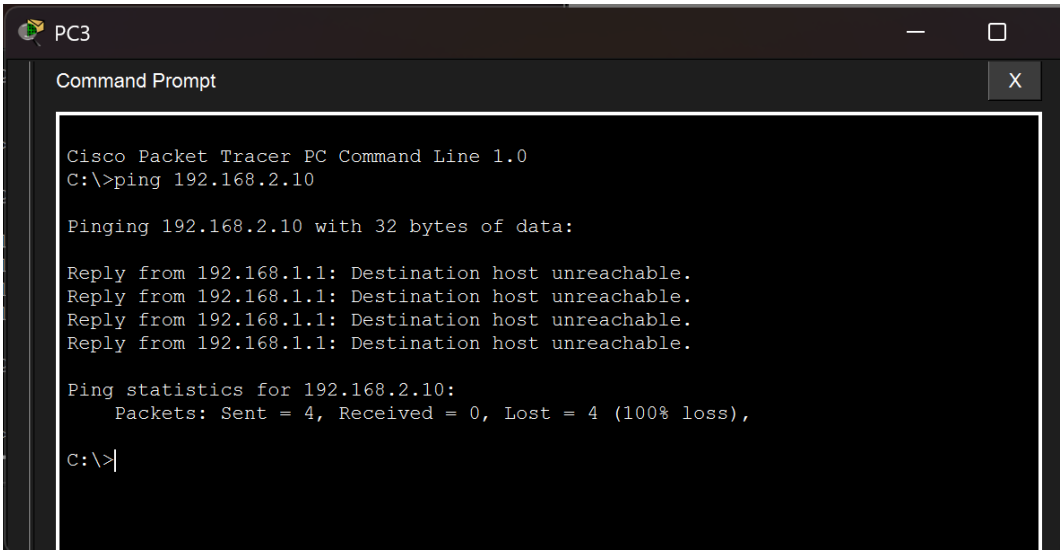
Router(config)#access-list 10 permit any
Router(config)#
```

Result:

From pc1 to pc2:



From pc3 to pc2:

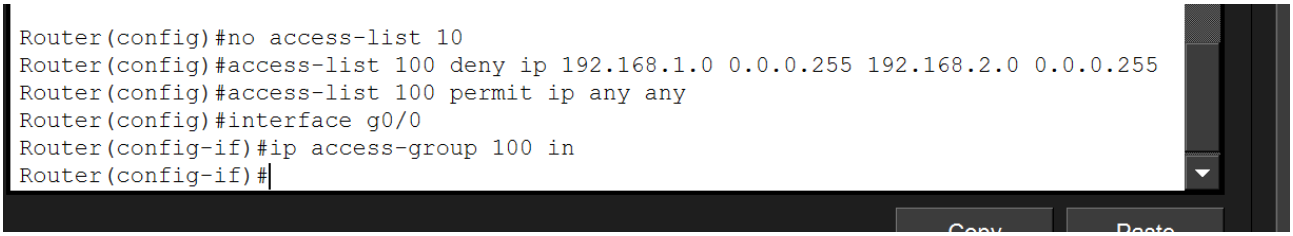


Stage 3 — Extended ACL (Source and Destination)

Objective

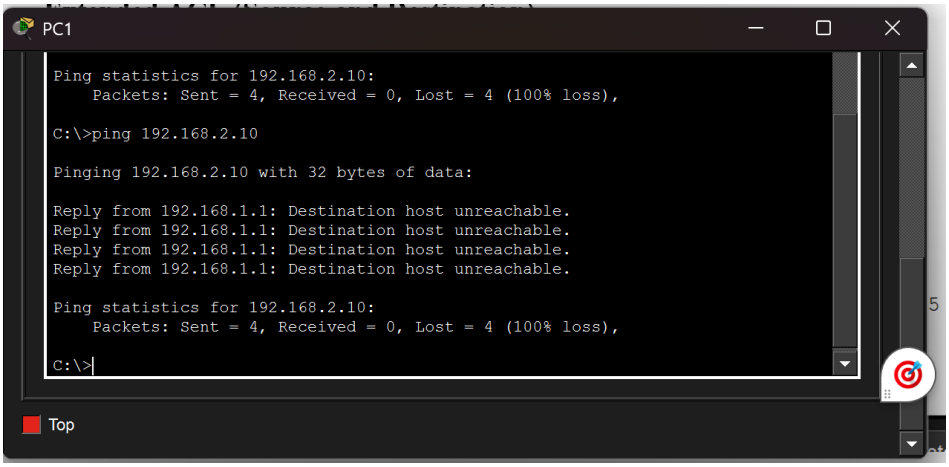
Block traffic only when:

- Source: 192.168.1.0/24
- Destination: 192.168.2.0/24

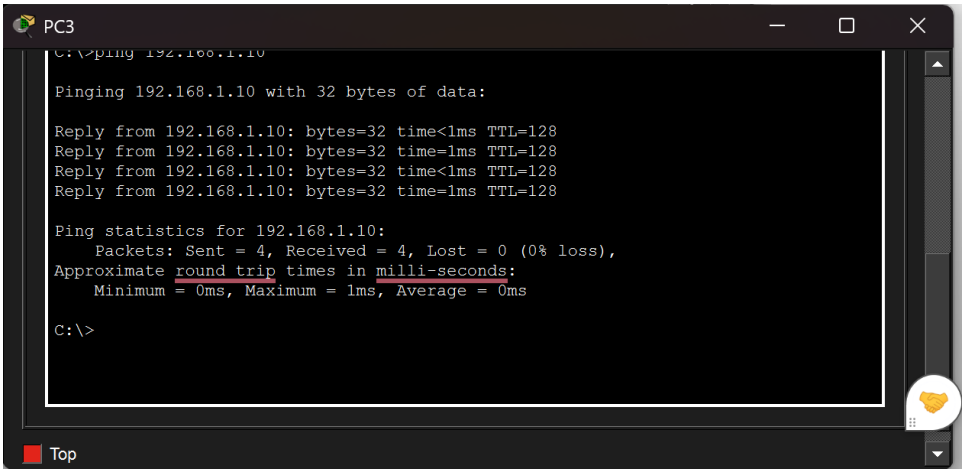


Result:

From pc1 to pc2:



From pc3 to pc1(in the same network):

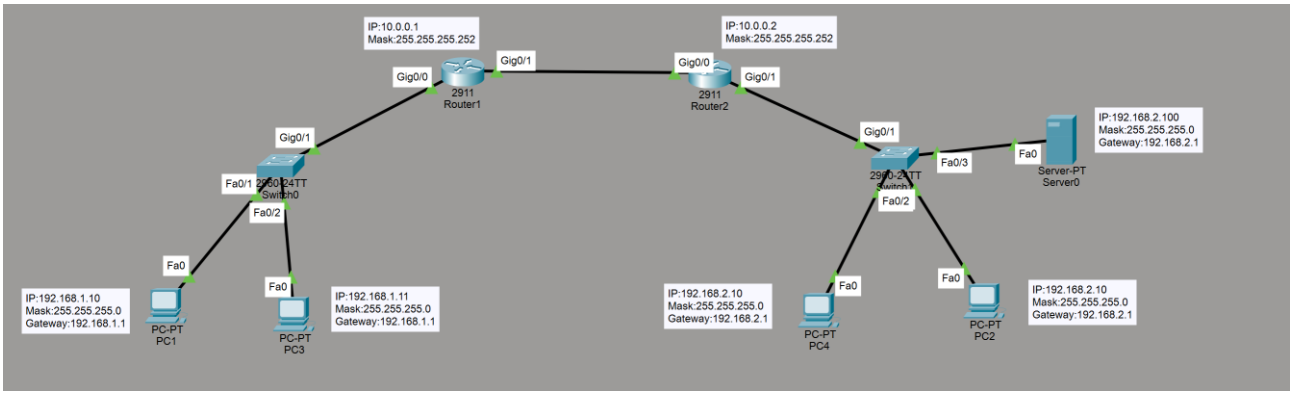


Stage 4 — HTTP Server Setup and Testing

Objective:

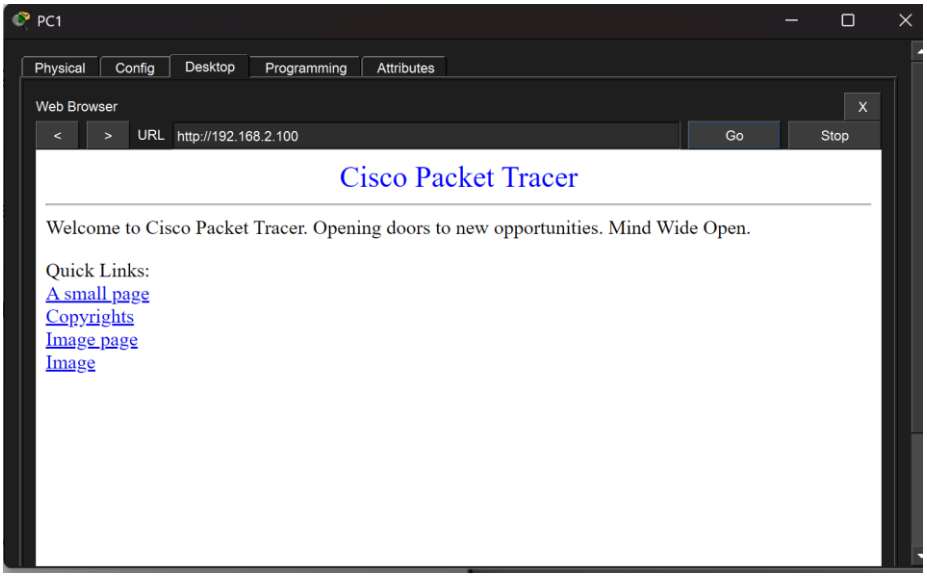
Test application-layer filtering using HTTP

Adding a server:



Result:

Check from pc1:



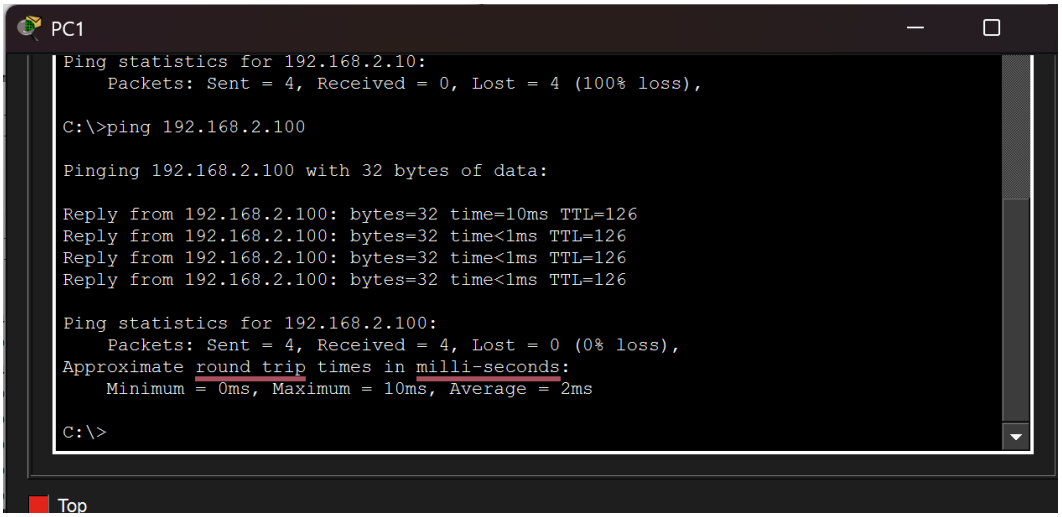
Stage 5 — Port-Based Filtering (HTTP)

Objective:

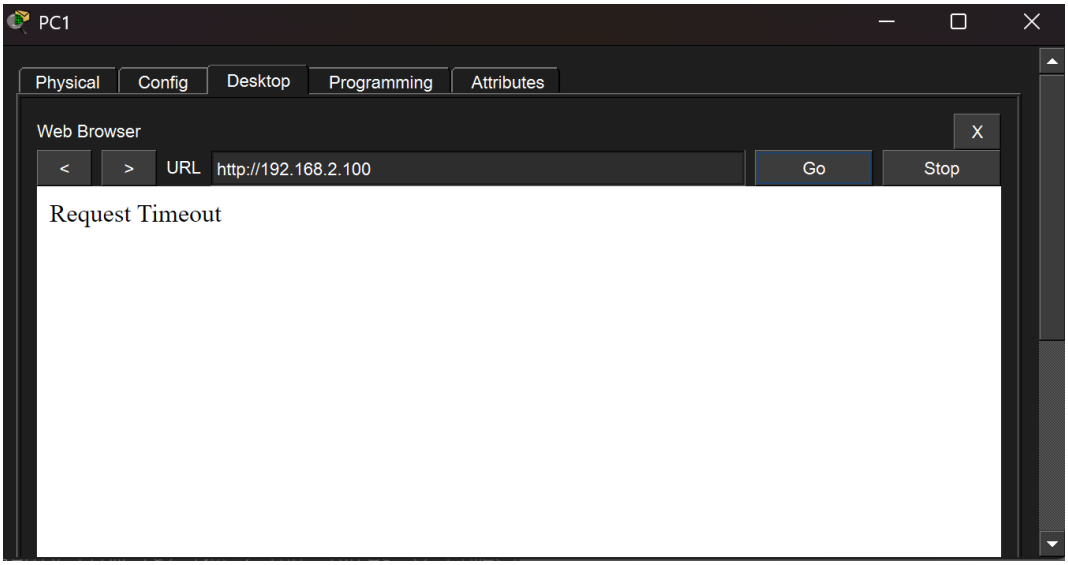
Block HTTP traffic only

```
Router#show access-lists
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#access-list 100 deny tcp 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255 eq 80
Router(config)#access-list 100 permit ip any any
Router(config)#interface g0/0
Router(config-if)#ip access-group 100 in
Router(config-if)#
```

Ping working:



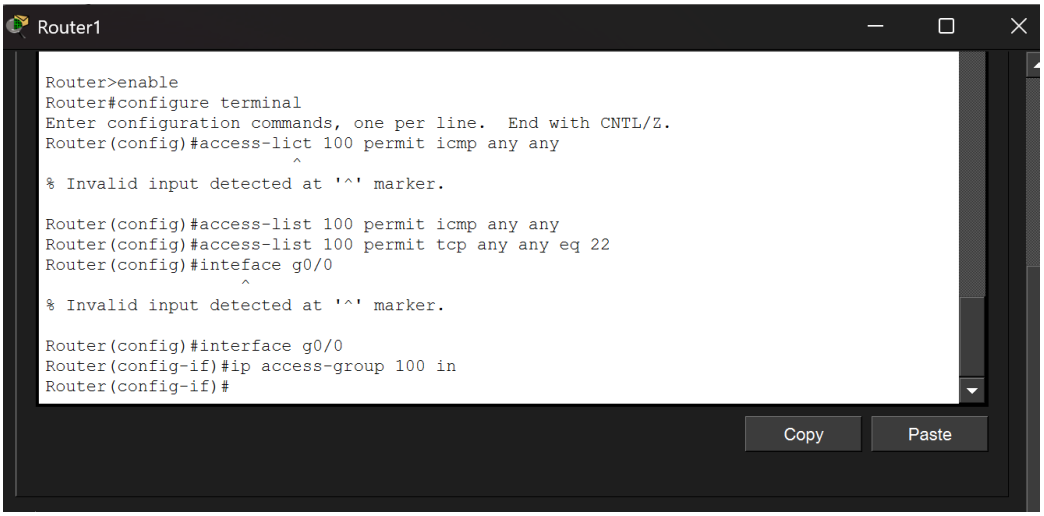
HTTP blocked:



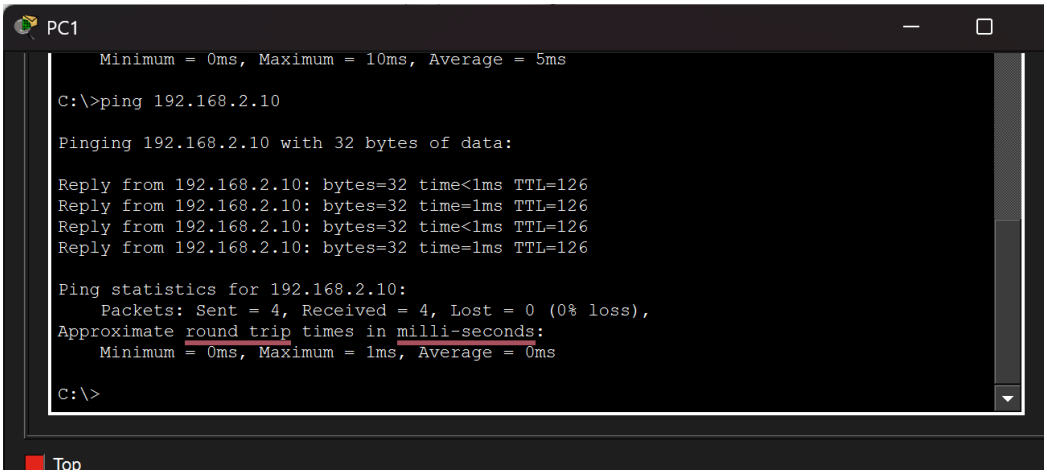
Stage 6 — Protocol Filtering

Objective:

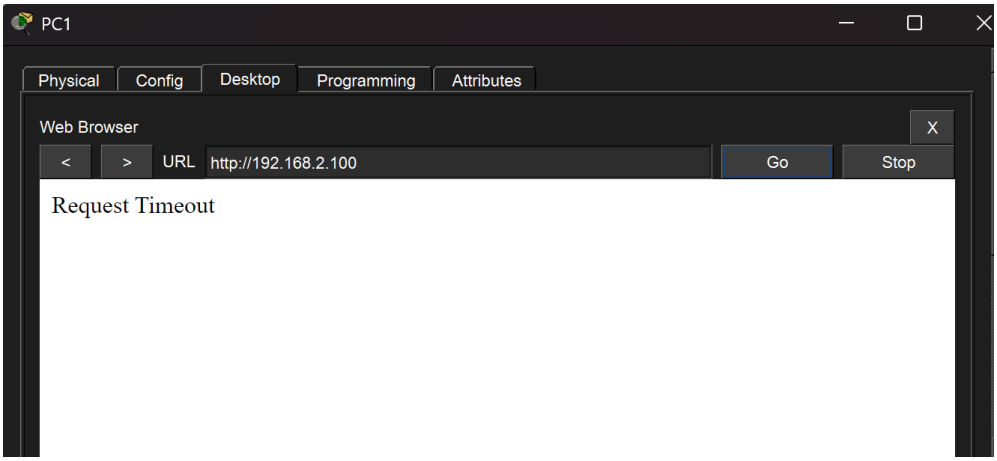
Allow only ICMP and SSH



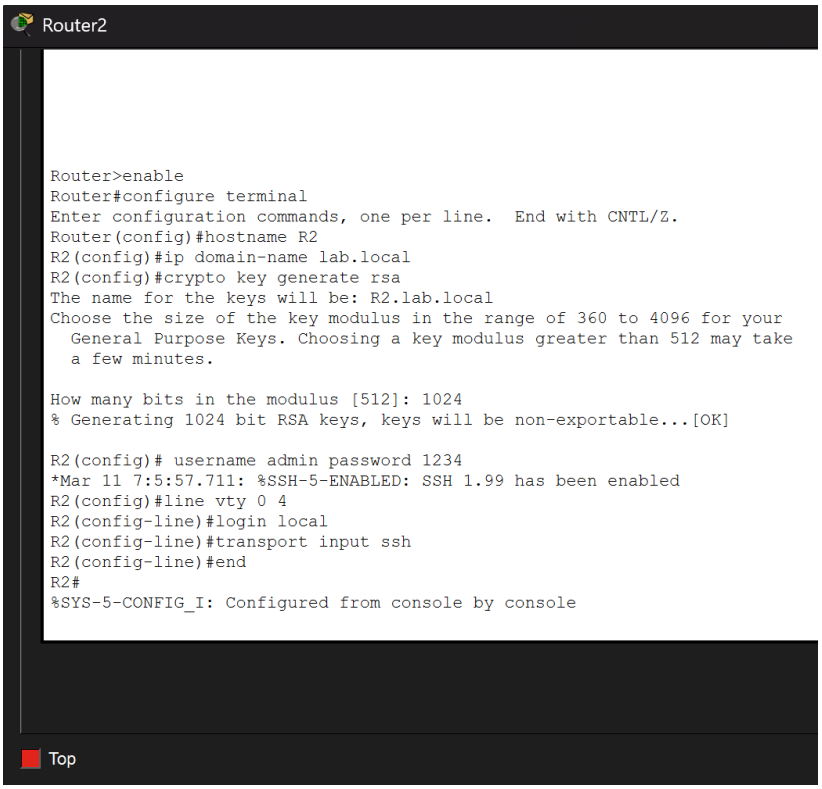
Ping working:



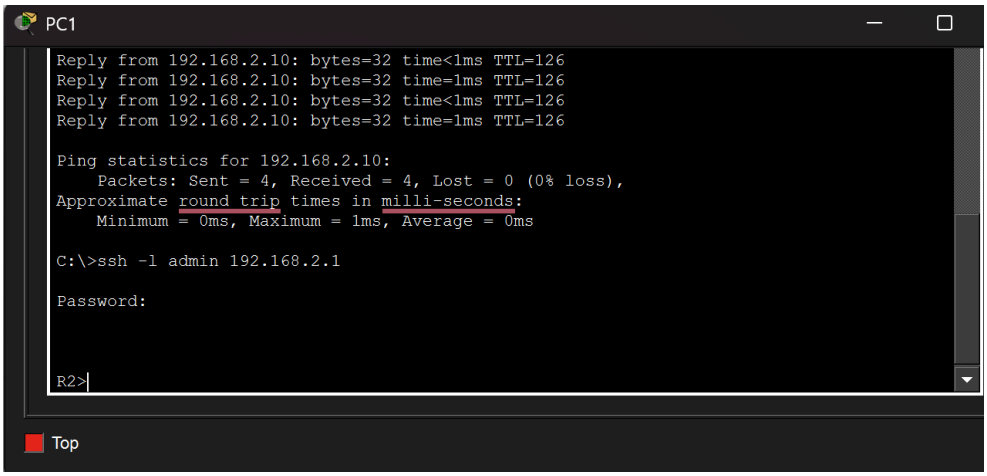
HTTP is not working:



SSH configured:



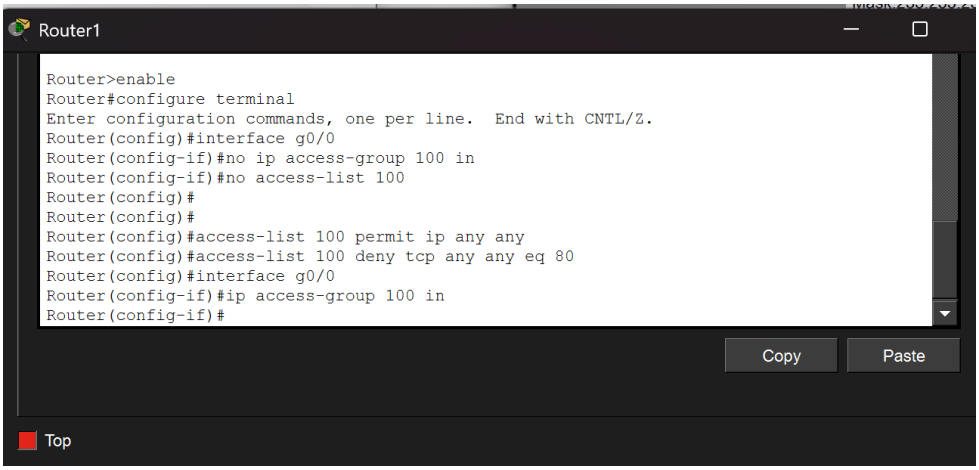
SSH working:



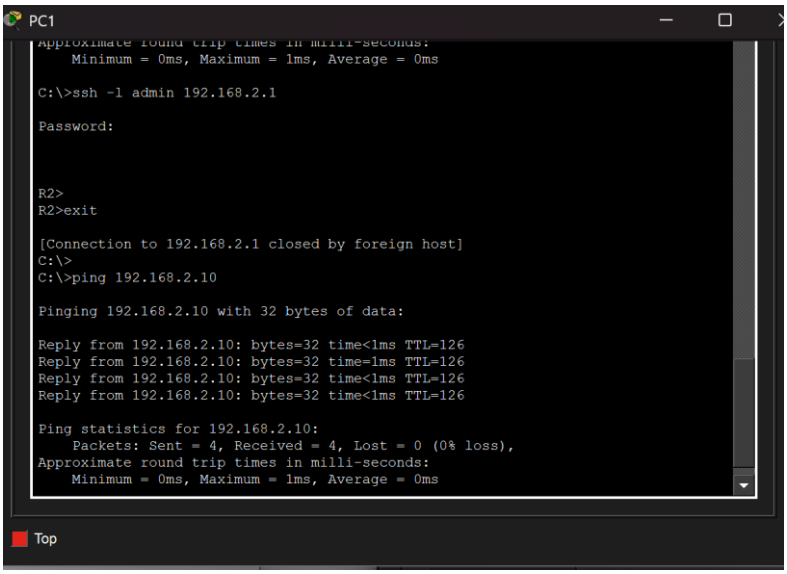
Stage 7 — Rule Order Experiment

Objective:

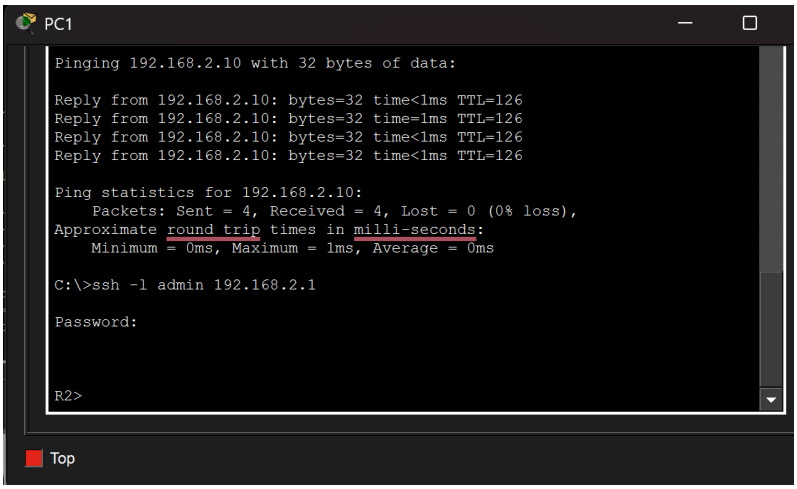
Understand rule evaluation order



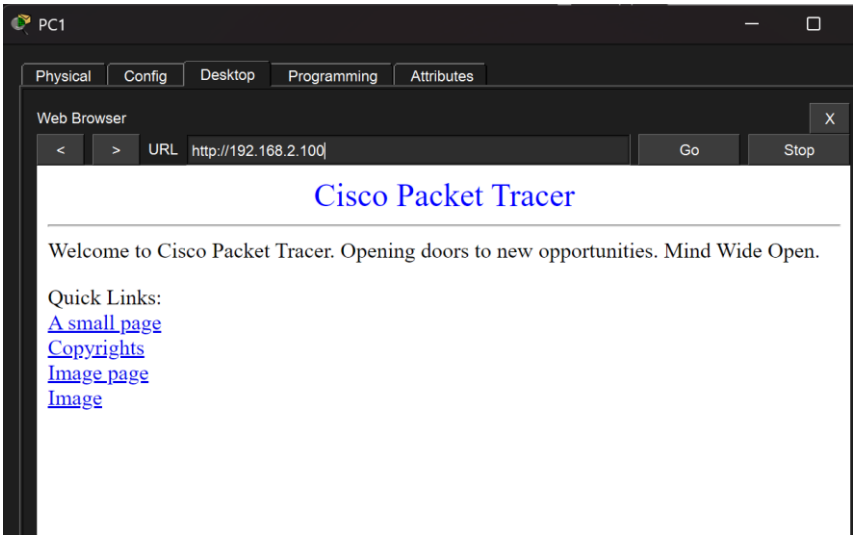
Ping is working:



SSH is working:



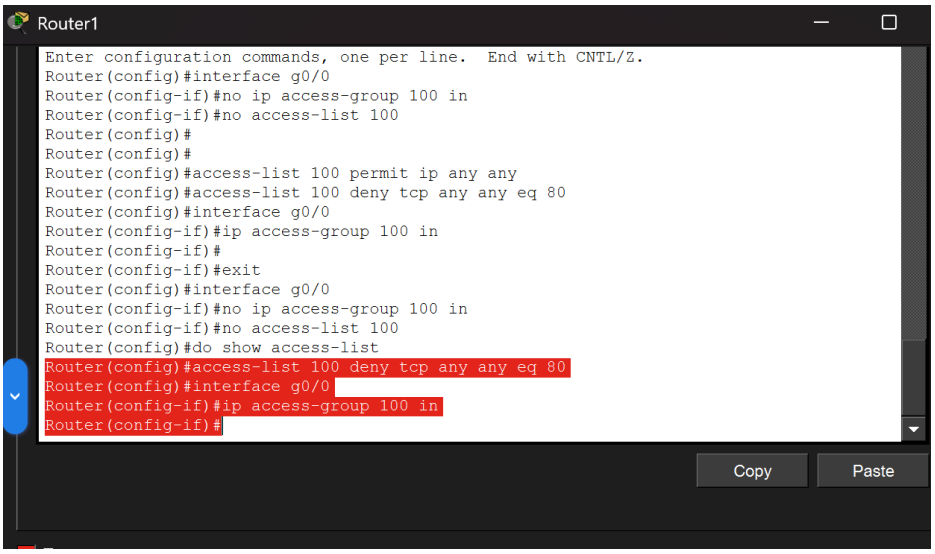
HTTP is working:



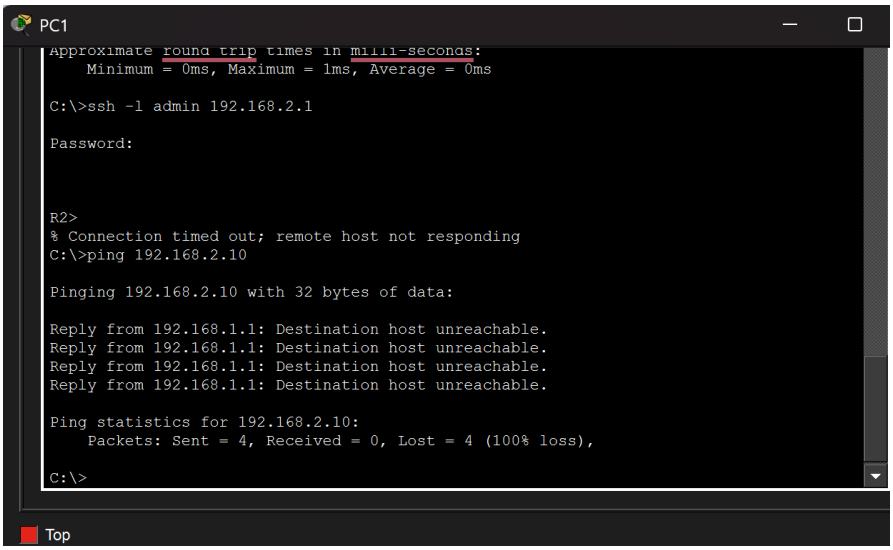
Stage 8 — Implicit Deny

Objective:

Observe default deny behavior



Ping is not working:

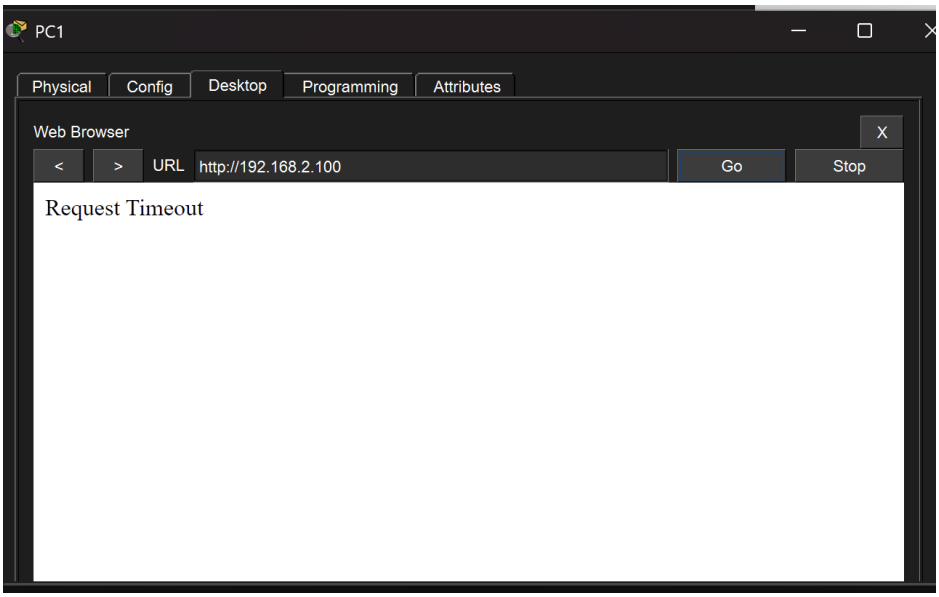


SSH is not working:

```
C:\>ssh -l admin 192.168.2.1

% Connection timed out; remote host not responding
C:\>
```

HTTP is not working:

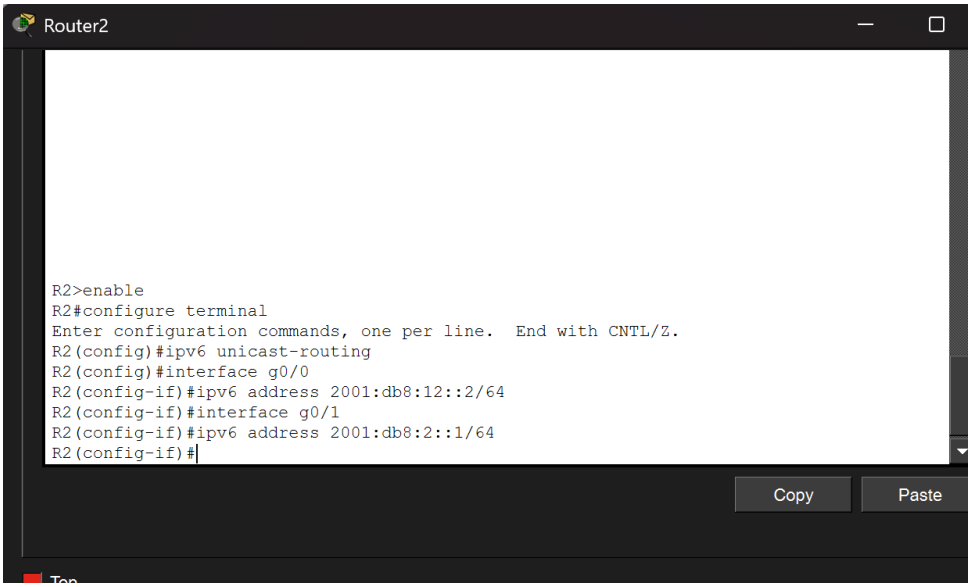


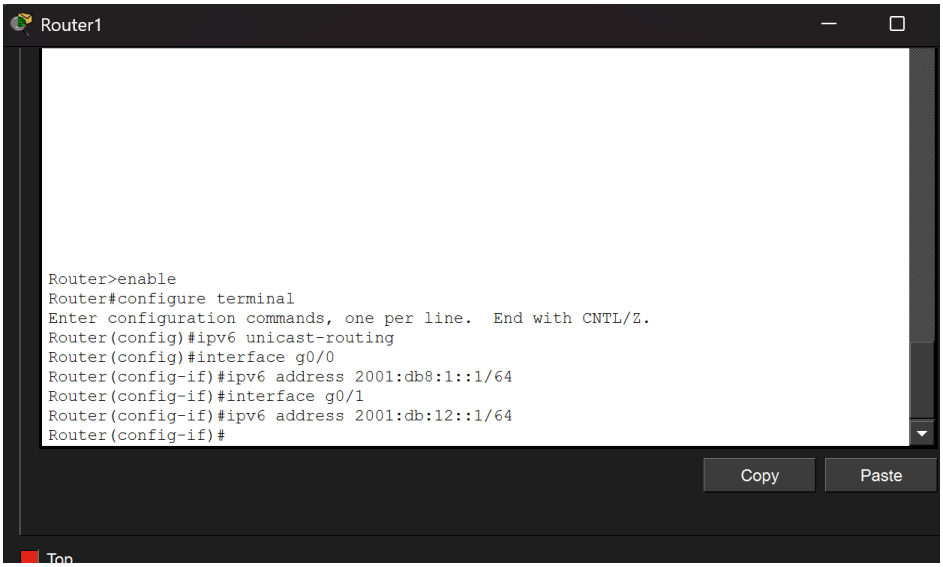
Stage 9 — IPv6 ACL

Objective:

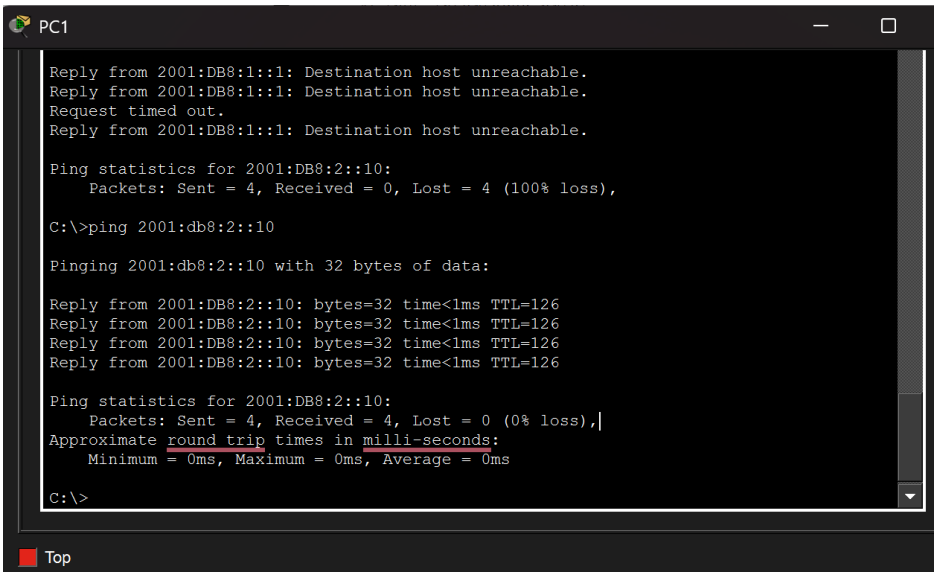
Apply ACL concepts using IPv6

Assigning IPv6 addresses:

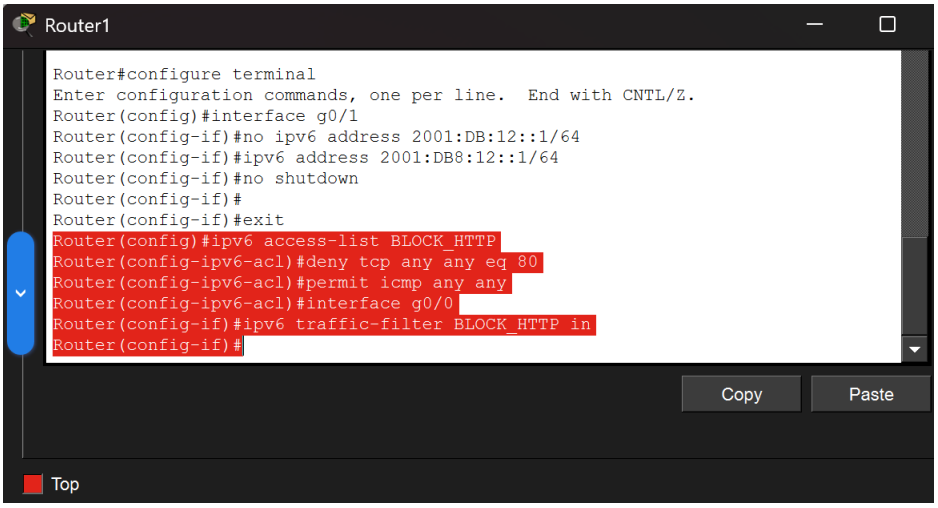




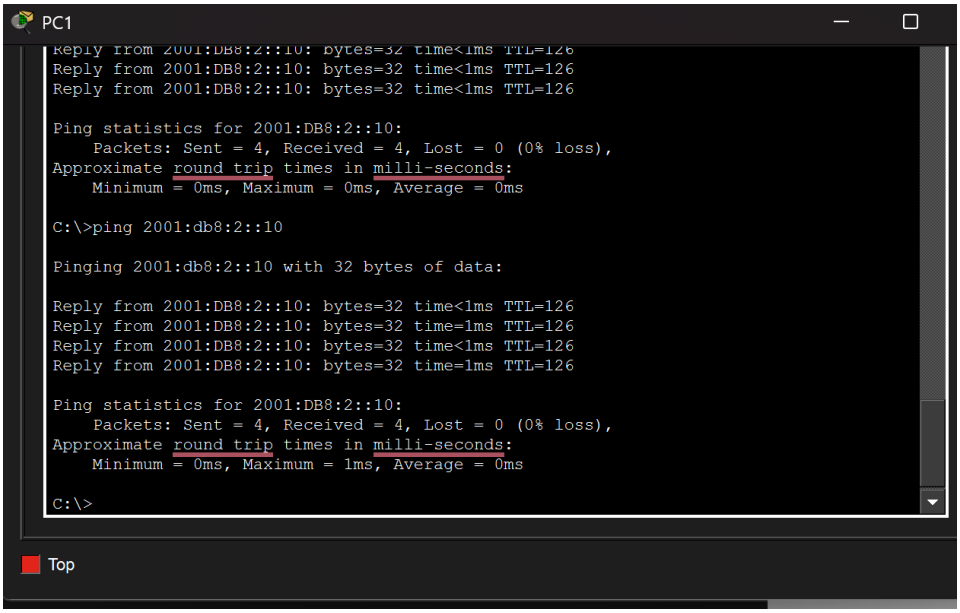
IPv6 ping working:



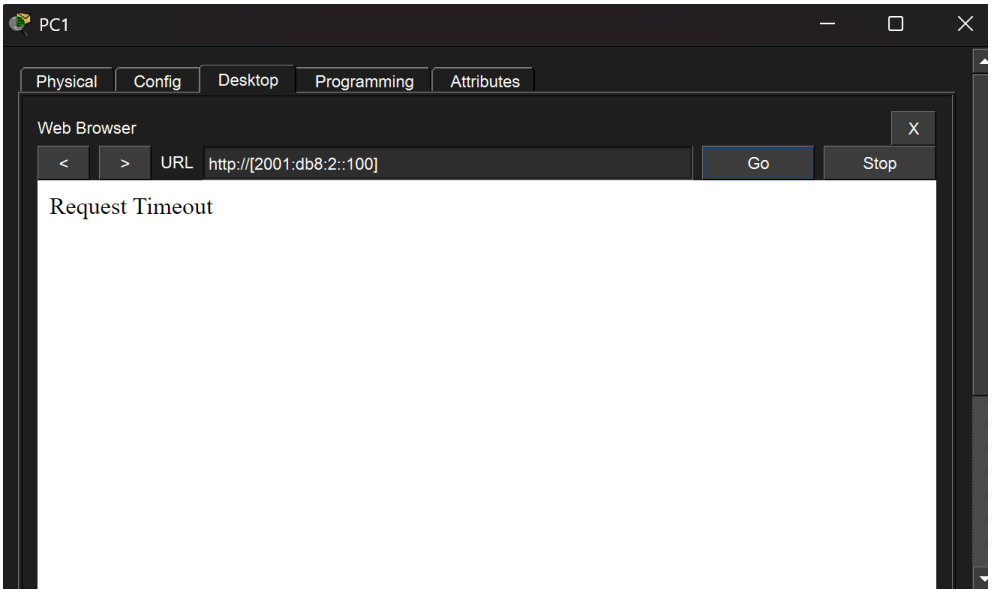
ACL:



Ping is working:

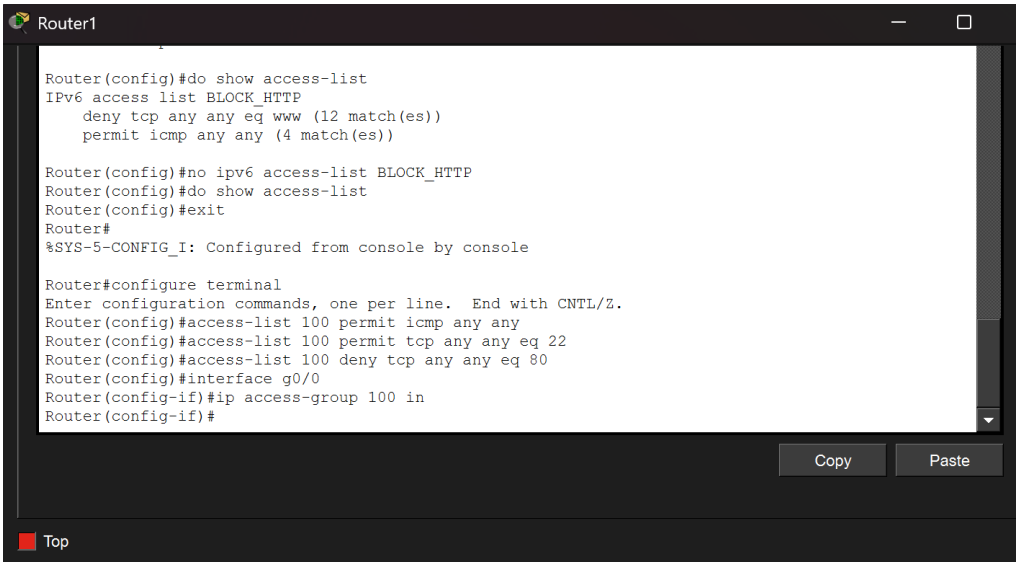


HTTP fails:

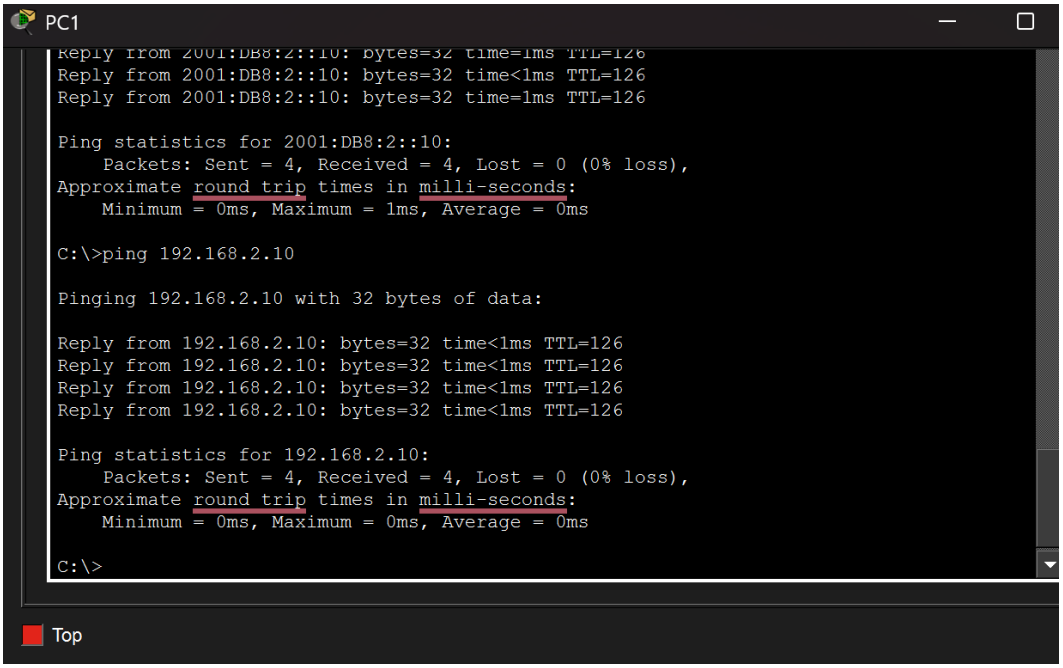


Final Task:

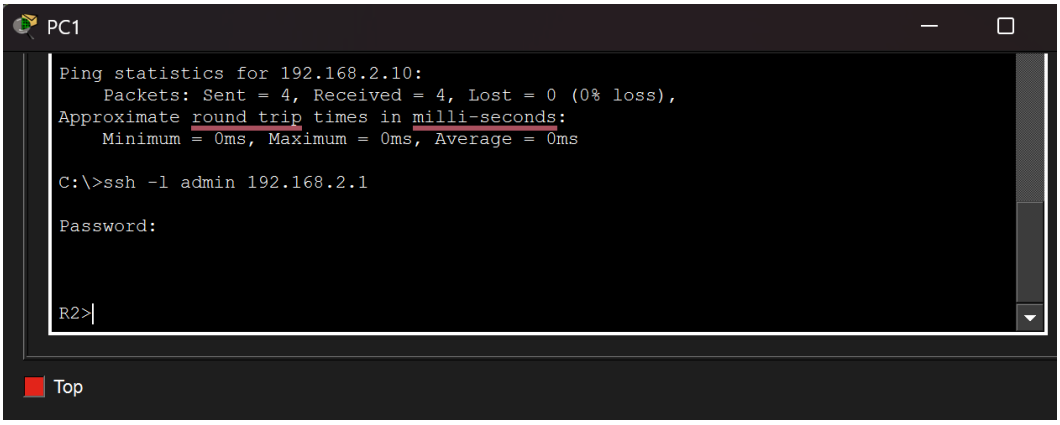
ACL:



Ping is working:



SSH is working:



HTTP is not working:

